

MACHINE LEARNING MIDSEM

Last-minute PDF cheat sheet from 2023–2026 regular, makeup, sample papers, and answer keys

Proper math edition: formulas render as real symbols, fractions, sums, Greek letters, and matrices.

How To Use This PDF

Start with the triage list. Then memorize the formulas and answer frames. The past papers reward step-by-step calculations plus one clear justification. Do not write only the final number.

Top revision order

1. Confusion matrix metrics: precision, recall, F1, specificity, cost, and threshold.
2. ID3 decision trees: entropy, weighted entropy, information gain, root choice.
3. Bias-variance and regularization: Ridge, Lasso, early stopping, train-test gaps.
4. Logistic regression: sigmoid, log-odds, large weights, threshold effects.
5. Data quality: missing values, duplicates, outliers, invalid formats, mixed units.
6. Feature scaling and gradient descent: z-score, high learning rate, normal equation cost.

Page map

Start here	Triage list and repeated paper pattern.
Basics + data	Learning types, Tom Mitchell E, T, P , data-quality issue bank.
Linear models	Regression formulas, gradient descent, scaling, bias-variance, Ridge/Lasso.
Classification	Logistic regression, log loss, thresholds, confusion-matrix metrics.
Trees + finish	Entropy, information gain, noise/outliers, ready-to-write answers, 20-minute drill.

Repeated Paper Pattern

Zone	Likely task	Fast answer target
Introduction	Learning type, Tom Mitchell E, T, P .	Classification: accuracy, precision, recall, F1, AUC. Regression: MSE, RMSE, R^2 .
ML workflow	Data quality, missing values, scaling, split strategy.	Name the issue, why it hurts, and the fix. Use stratified split for imbalance.
Regression	Linear regression, dimensions, GD, RMSE, R^2 , regularization.	For huge data prefer gradient descent; matrix inversion is expensive.
Classification	Logistic regression, sigmoid, log-odds, threshold.	Weights affect log-odds; threshold changes TP, FP, FN, TN.
Evaluation	Confusion matrix and metric choice.	Accuracy can lie under imbalance. Choose metric based on cost.
Decision tree	Entropy, gain, majority vote, root node.	Choose maximum gain or minimum weighted entropy. Do not evaluate only on training data.

Core Basics

Tom Mitchell Definition

A program learns from **experience** E , for **task** T , using **performance measure** P , if performance on T , as measured by P , improves with E .

Weather example

- E : historical weather data.
- T : classify weather as Sunny, Windy, or Rainy.
- P : accuracy, precision, recall, F1, AUC, or misclassification rate.
- If target is rainfall amount, it is regression. P can be MSE, RMSE, or R^2 .

Learning Type Decoder

Type	Exam wording
Supervised classification	Labels are given; target is a class, such as positive/negative or approved/rejected.
Supervised regression	Labels are given; target is a continuous number, such as rainfall or house price.
Unsupervised	No labels; find groups, clusters, or hidden structure.
Reinforcement	Agent acts, gets rewards, and improves by interaction.
Generative classifier	Use when density $P(x y)$ is needed, such as outlier detection.

Data Quality: Easy Marks

Issue	What it looks like	Fix
Missing value	Blank email, height, course ID, feature value.	Mean/median for numeric; mode or Unknown for categorical.
Duplicate row	Same person appears twice, sometimes with different target.	Remove exact duplicates or resolve conflict using stable keys.
Outlier	Age 350, age 58, CGPA 87.5, extreme area.	Validate range, investigate, cap, transform, or use robust imputation.
Wrong unit	kg mixed with lbs; square metres mixed with square feet.	Convert all values to one unit.
Bad format	Date formats differ; email written as dev[at]mail.com.	Standardize with parser, regex, or validation rules.
Invalid category	Bad blood group, AI vs Artificial Intelligence.	Use allowed-value checks and mapping tables.
Label mismatch	1/0 mixed with Positive/Negative.	Map labels to one representation before training.

Write this in the exam

Problem: The value ____ is invalid, inconsistent, missing, duplicated, or an outlier.

Why it hurts: It can mislead training or evaluation.

Fix: Apply ____ because ____.

Regression + Gradient Descent

Linear Regression

Prediction

$$\begin{aligned}\hat{y} &= \theta_0 + \theta_1 x_1 + \theta_2 x_2 + \cdots + \theta_n x_n \\ &= \boldsymbol{\theta}^\top \mathbf{x}\end{aligned}$$

θ_0 is the intercept. Each θ_j is the change in prediction for one-unit change in feature x_j , keeping other features fixed. If area changes from m^2 to ft^2 , the coefficient rescales. The relationship need not change.

MSE, RMSE, and R^2

Regression metrics

$$\begin{aligned}\text{MSE} &= \frac{1}{m} \sum_{i=1}^m \left(\hat{y}^{(i)} - y^{(i)} \right)^2 \\ \text{RMSE} &= \sqrt{\text{MSE}} \\ R^2 &= 1 - \frac{\sum_i (y^{(i)} - \hat{y}^{(i)})^2}{\sum_i (y^{(i)} - \bar{y})^2}\end{aligned}$$

R^2 is the proportion of variance in the target explained by the model. Higher test R^2 is better, but still check RMSE and overfitting.

Gradient Descent

Update rule

$$\theta_j := \theta_j - \alpha \frac{\partial J(\boldsymbol{\theta})}{\partial \theta_j}$$

α is the learning rate. If $J(\boldsymbol{\theta})$ increases over iterations, α is probably too large. Decrease it.

Gradient interpretation

If $\frac{\partial J}{\partial \theta_1}$ is large, changing θ_1 can still reduce error a lot. If $\frac{\partial J}{\partial \theta_2} \approx 0$, then θ_2 is near an optimum or the feature is weak.

Feature Scaling

Z-score normalization

$$z = \frac{x - \mu}{\sigma}$$

μ is the mean and σ is the standard deviation. Scaling makes features comparable and helps gradient descent converge faster. It does not guarantee a global optimum and it does not fix a singular $X^\top X$.

Normal Equation Dimensions

Closed form

$$\boldsymbol{\theta} = (X^\top X)^{-1} X^\top \mathbf{y}$$

If $m = 28$ examples and $n = 4$ features, add an intercept column. Then $X \in \mathbb{R}^{28 \times 5}$, $\mathbf{y} \in \mathbb{R}^{28 \times 1}$, and $\boldsymbol{\theta} \in \mathbb{R}^{5 \times 1}$. For huge data, prefer gradient descent because matrix inversion is expensive.

Bias, Variance, and Regularization

Training error	Validation error	Diagnosis	Fix
High	High and close	High bias / underfitting	Add features, reduce λ , use richer model.
Low	High, large gap	High variance / overfitting	Regularize, simplify, early stop, prune, or get cleaner data.
Low	Low and close	Good fit	Keep the test set untouched until final evaluation.

Ridge and Lasso

Regularized objectives

$$J_{\text{ridge}}(\boldsymbol{\theta}) = \text{MSE} + \lambda \sum_{j=1}^n \theta_j^2$$

$$J_{\text{lasso}}(\boldsymbol{\theta}) = \text{MSE} + \lambda \sum_{j=1}^n |\theta_j|$$

- **Ridge / L2:** shrinks all weights and usually keeps all features. Larger λ increases bias and reduces variance.
- **Lasso / L1:** can set coefficients exactly to zero. Use it when many predictors are irrelevant.
- **Early stopping:** stop when validation loss starts rising while training loss keeps falling.

Common exam trap

If training and validation errors are both high and close, do *not* say high variance. Say high bias / underfitting. In Ridge, reduce λ to reduce bias.

Logistic Regression

Sigmoid and Log-Odds

Core equations

$$z = \boldsymbol{\theta}^\top \mathbf{x}$$

$$h_{\boldsymbol{\theta}}(\mathbf{x}) = \sigma(z) = \frac{1}{1 + e^{-z}}$$

$$\log\left(\frac{p}{1-p}\right) = \boldsymbol{\theta}^\top \mathbf{x}$$

A coefficient changes **log-odds**, not the final decision alone. A large positive CGPA weight does not guarantee selection. Other weights and θ_0 also affect $\boldsymbol{\theta}^\top \mathbf{x}$.

Cross-Entropy Loss

Log loss

$$J(\boldsymbol{\theta}) = -\frac{1}{m} \sum_{i=1}^m \left[y^{(i)} \log h_{\boldsymbol{\theta}}(x^{(i)}) + (1 - y^{(i)}) \log (1 - h_{\boldsymbol{\theta}}(x^{(i)})) \right]$$

One Batch Gradient Descent Update

Logistic gradient

$$\frac{\partial J}{\partial \theta_j} = \frac{1}{m} \sum_{i=1}^m \left(h_{\boldsymbol{\theta}}(x^{(i)}) - y^{(i)} \right) x_j^{(i)}$$

$$\theta_j^{\text{new}} = \theta_j^{\text{old}} - \alpha \frac{\partial J}{\partial \theta_j}$$

Large weights and overfitting

- $w = 1$: smooth sigmoid.
- $w = 10$: steep sigmoid.
- $w = 100$: almost a step function.
- Large weights make tiny input changes cause huge probability jumps. This can memorize training quirks.

Classification Metrics

Use the positive class unless the question asks for the negative class. If it asks for the negative class, swap the view: TN behaves like the “TP” of the negative class.

Confusion-matrix formulas

$$\text{Accuracy} = \frac{TP + TN}{TP + FP + FN + TN}$$

$$\text{Precision} = \frac{TP}{TP + FP}$$

$$\text{Recall} = \text{Sensitivity} = \frac{TP}{TP + FN}$$

$$F_1 = \frac{2PR}{P + R}$$

$$\text{Specificity} = \frac{TN}{TN + FP}$$

Metric	Meaning	Use when
Precision	Among predicted positives, how many were correct?	False positives are costly.
Recall	Among actual positives, how many did we catch?	False negatives are costly.
F1	Balance between precision and recall.	Both error types matter.
Specificity	Among actual negatives, how many were correctly rejected?	Need to avoid false alarms.
ROC-AUC	Threshold-independent ranking quality.	Comparing classifiers under imbalance.

Rare disease threshold answer

If false negatives are much costlier than false positives, lower the threshold. This increases recall and catches more diseased patients. Precision may decrease because more healthy patients are flagged.

Worked Cost Pattern

If $FN = 30$, $FP = 85$, cost of FN is 50,000, and cost of FP is 2,000, then

Cost of errors

$$\begin{aligned}
 \text{Cost} &= 30(50,000) + 85(2,000) \\
 &= 1,500,000 + 170,000 \\
 &= 1,670,000.
 \end{aligned}$$

Since false negatives dominate the cost, prioritize recall.

Decision Trees: Entropy + ID3

Entropy

Entropy

$$H(S) = - \sum_{i=1}^k p_i \log_2(p_i)$$

Entropy is impurity. A pure node has $H(S) = 0$. For four uniform classes, entropy is $\log_2 4 = 2$ bits. A rarer event carries more information:

Information in one event

$$I(x) = -\log_2 p(x)$$

So $p = 0.2$ needs more bits than $p = 0.8$.

Information Gain

ID3 split score

$$\text{Gain}(S, A) = H(S) - \sum_{v \in \text{Values}(A)} \frac{|S_v|}{|S|} H(S_v)$$

Choose the attribute with maximum information gain. Equivalently, choose the attribute with minimum weighted child entropy.

Worked mini-example

Parent: 9 Repay, 5 Default.

$$H(S) = -\frac{9}{14} \log_2 \frac{9}{14} - \frac{5}{14} \log_2 \frac{5}{14} \approx 0.94.$$

High branch: 6 Repay, 2 Default.

$$H_{\text{High}} = -\frac{6}{8} \log_2 \frac{6}{8} - \frac{2}{8} \log_2 \frac{2}{8} \approx 0.81.$$

Low branch: 3 Repay, 3 Default.

$$H_{\text{Low}} = -\frac{3}{6} \log_2 \frac{3}{6} - \frac{3}{6} \log_2 \frac{3}{6} = 1.$$

Weighted entropy:

$$H_{\text{after}} = \frac{8}{14}(0.81) + \frac{6}{14}(1.00) \approx 0.8915.$$

Information gain:

$$\text{Gain} = 0.94 - 0.8915 \approx 0.0485.$$

Conclusion: low gain, so Income Level is not a strong split in this example.

Information gain limitation

Information gain can favor attributes with many unique values, such as ID numbers. Gain ratio in C4.5 reduces this high-cardinality bias.

Noise, Outliers, and Splits

Scenario	Linear regression	Logistic regression	Decision tree
Noisy irrelevant features	Affected. Noise can get non-zero weights.	Affected. Noise can distort the decision boundary.	Strongly affected. Greedy splits can fit random quirks.
Outliers	Strongly affected under squared error.	Less affected, but not immune; sigmoid saturates.	Moderately affected; may create special branches.
Imbalanced classes	Use suitable regression metric if target is numeric.	Accuracy can mislead; use precision, recall, F1, AUC.	Use stratified split and class-aware metrics.

Train-validation-test rule

Use training data to fit parameters. Use validation data to tune model choices. Use test data once for final unseen evaluation. For imbalanced data, use stratified splitting to preserve class proportions.

Ready-To-Write Answer Frames

High bias / underfit

Training and validation errors are both high and close. The model is too simple, so it misses real patterns. Reduce regularization, add useful features, or use a richer model.

High variance / overfit

Training error is low but validation error is high. The model learned noise. Use regularization, early stopping, a simpler model, pruning, or more clean data.

Threshold changes

Changing the threshold changes predicted labels. That changes TP, FP, FN, and TN. Therefore precision, recall, specificity, and F1 can change.

Feature scaling

Scaling makes features comparable. Gradient descent reaches a good solution faster because the cost surface is less stretched. It does not guarantee the global minimum.

Decision tree evaluation

A deep tree can memorize training data and show near 100% training accuracy. That does not prove generalization. Use validation or unseen test data.

Generative classifier

Choose generative when the task needs $P(x | y)$, such as density estimation or outlier detection. Discriminative models mainly learn the boundary or $P(y | x)$.

20-Minute Drill

1. 4 min: Recite metric formulas. Then compute one confusion matrix by hand.
2. 4 min: Do one entropy problem: parent, children, weighted entropy, gain, conclusion.
3. 3 min: Write Ridge and Lasso objectives from memory.
4. 3 min: Explain sigmoid weights $w = 1, 10, 100$ and why large weights overfit.
5. 3 min: List 8 data-quality issues and one fix for each.
6. 3 min: Practice one high-bias answer and one high-variance answer.